INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – I ,Third Semester Examination, August- 2025

Subject Code: BCACC302

Subject Name: Operating Systems

Full Marks: 30 Duration: 1 Hour

Part - A
Attempt 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Define Operating System. **2**
2. Define Multiprocessing. **2**
3. What is Process? **2**
4. Define Job Queue. **2**
5. Explain Synchronization. **2**
6. Define Race Condition **2**

Part - B
Attempt 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Describe Process Control Block. **5**
8. Discuss FCFS Algorithm. **5**
9. Compare and contrast multitasking, multiprocessing, and multithreading. **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Consider the following table and evaluate the average turnaround time and average waiting time for priority scheduling algorithm for both FCFS and SJF cases.

Process	Burst Time	Arrival Time
A	10	3
B	5	6
C	8	0

10

11. Discuss the role of semaphores in solving the critical section problem. Write the difference between Binary and Counting semaphore with an example.

10

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